# IM POLINY OSIPENKO, Russian Federation

WMO: **314160**

Lat: **52.4207N** Lon: **136.4799E** Elev: **70** StdP: **100.48** Time Zone: **10.00 (E10)** Period: **94-19** WBAN: **99999**

## Annual Heating, Humidification, and Ventilation Design Conditions

| | Coldest Month | Heating DB | | Humidification DP/MCDB and HR | | | | | | Coldest Month WS/MCDB | | | | MCWS/PCWD to 99.6% DB | | WSF |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 99.6% | | | 99% | | | 0.4% | | 1% | | | | |
| | | 99.6% | 99% | DP | HR | MCDB | DP | HR | MCDB | WS | MCDB | WS | MCDB | MCWS | PCWD | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) |
| (1) | **1** | **-39.9** | **-37.3** | **-42.2** | **0.1** | **-39.5** | **-39.8** | **0.1** | **-37.1** | **9.7** | **-11.0** | **7.5** | **-11.5** | **0.2** | **0** | **0.599** |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| | Hottest Month | Hottest Month DB Range | Cooling DB/MCWB | | | | | | Evaporation WB/MCDB | | | | | | MCWS/PCWD to 0.4% DB | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 0.4% | | 1% | | 2% | | 0.4% | | 1% | | 2% | | | |
| | | | DB | MCWB | DB | MCWB | DB | MCWB | WB | MCDB | WB | MCDB | WB | MCDB | MCWS | PCWD |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) |
| (2) | **7** | **10.4** | **29.6** | **21.1** | **27.7** | **20.1** | **25.9** | **19.0** | **22.5** | **27.8** | **21.2** | **26.1** | **20.1** | **24.5** | **1.9** | **180** |

| | Dehumidification DP/MCDB and HR | | | | | | | | | Enthalpy/MCDB | | | | | | Extreme Max WB |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0.4% | | | 1% | | | 2% | | | 0.4% | | 1% | | 2% | | |
| | DP | HR | MCDB | DP | HR | MCDB | DP | HR | MCDB | Enth | MCDB | Enth | MCDB | Enth | MCDB | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) |
| (3) | **20.7** | **15.5** | **25.5** | **19.5** | **14.3** | **24.2** | **18.3** | **13.3** | **22.9** | **67.0** | **27.7** | **62.3** | **26.1** | **58.2** | **24.6** | **29.4** |

## Extreme Annual Design Conditions

| | Extreme Annual WS | | | | Extreme Annual Temperature | | | | n-Year Return Period Values of Extreme Temperature | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Mean | | Standard Deviation | | n=5 years | | n=10 years | | n=20 years | | n=50 years | |
| | 1% | 2.5% | 5% | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max |
| | (a) | (b) | (c) | | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) |
| (4) | **6.9** | **5.7** | **4.9** | DB | **-42.5** | **32.7** | **2.8** | **2.0** | **-44.5** | **34.2** | **-46.2** | **35.3** | **-47.8** | **36.4** | **-49.8** | **37.9** |
| (5) | | | | WB | **-42.5** | **24.3** | **2.7** | **1.6** | **-44.4** | **25.5** | **-45.9** | **26.4** | **-47.4** | **27.3** | **-49.4** | **28.5** |

## Monthly Climatic Design Conditions

| | | | | Annual | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) |
| (6) | Temperatures, Degree-Days and Degree-Hours | | DBAvg | **-1.3** | **-25.0** | **-20.3** | **-10.4** | **0.8** | **8.8** | **15.4** | **18.9** | **17.7** | **11.6** | **2.0** | **-12.2** | **-23.6** |
| (7) | | | DBStd | **16.53** | **6.45** | **5.62** | **5.57** | **4.55** | **4.63** | **4.56** | **3.86** | **3.26** | **3.83** | **4.28** | **7.15** | **6.63** |
| (8) | | | HDD10.0 | **4916** | **1086** | **849** | **632** | **277** | **79** | **7** | **0** | **0** | **27** | **251** | **666** | **1043** |
| (9) | | | HDD18.3 | **7271** | **1344** | **1082** | **890** | **525** | **296** | **111** | **40** | **52** | **204** | **508** | **916** | **1302** |
| (10) | | | CDD10.0 | **803** | **0** | **0** | **0** | **2** | **43** | **168** | **276** | **239** | **73** | **1** | **0** | **0** |
| (11) | | | CDD18.3 | **117** | **0** | **0** | **0** | **0** | **2** | **23** | **58** | **32** | **2** | **0** | **0** | **0** |
| (12) | | | CDH23.3 | **1092** | **0** | **0** | **0** | **1** | **38** | **267** | **527** | **246** | **14** | **0** | **0** | **0** |
| (13) | | | CDH26.7 | **286** | **0** | **0** | **0** | **0** | **9** | **68** | **166** | **44** | **0** | **0** | **0** | **0** |
| (14) | Wind | | WSAvg | **1.7** | **1.1** | **1.4** | **1.9** | **2.3** | **2.5** | **1.9** | **1.8** | **1.6** | **1.5** | **1.8** | **1.6** | **1.3** |
| (15) | Precipitation | | PrecAvg | **488** | **11** | **7** | **13** | **28** | **48** | **59** | **84** | **90** | **66** | **45** | **22** | **16** |
| (16) | | | PrecMax | **681** | **54** | **33** | **40** | **84** | **120** | **146** | **210** | **235** | **144** | **143** | **59** | **60** |
| (17) | | | PrecMin | **250** | **0** | **0** | **0** | **4** | **5** | **11** | **4** | **28** | **5** | **0** | **1** | **0** |
| (18) | | | PrecStd | **97** | **10** | **7** | **10** | **19** | **25** | **30** | **47** | **47** | **35** | **31** | **16** | **15** |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% | DB | | **-4.9** | **-3.6** | **6.2** | **19.1** | **27.8** | **31.1** | **33.1** | **30.0** | **25.2** | **17.0** | **4.2** | **-2.1** |
| (20) | | | MCWB | | **-5.7** | **-5.2** | **1.7** | **10.8** | **15.6** | **20.9** | **23.7** | **22.2** | **17.9** | **10.9** | **0.8** | **-2.9** |
| (21) | | 2% | DB | | **-8.6** | **-6.8** | **3.1** | **14.8** | **23.5** | **28.6** | **30.4** | **27.9** | **22.5** | **12.9** | **1.3** | **-6.4** |
| (22) | | | MCWB | | **-9.3** | **-8.2** | **-0.7** | **7.8** | **13.7** | **19.3** | **21.9** | **20.9** | **16.4** | **7.8** | **-0.7** | **-7.2** |
| (23) | | 5% | DB | | **-11.5** | **-9.1** | **0.7** | **11.7** | **21.0** | **26.4** | **28.4** | **26.1** | **20.4** | **10.5** | **-0.4** | **-10.0** |
| (24) | | | MCWB | | **-12.1** | **-10.3** | **-2.3** | **5.8** | **12.2** | **18.1** | **21.0** | **19.9** | **14.9** | **6.3** | **-1.8** | **-10.8** |
| (25) | | 10% | DB | | **-14.6** | **-11.2** | **-1.5** | **8.7** | **18.3** | **24.1** | **26.3** | **24.3** | **18.3** | **8.5** | **-2.0** | **-13.4** |
| (26) | | | MCWB | | **-15.2** | **-12.2** | **-3.9** | **3.6** | **11.0** | **16.8** | **19.9** | **18.7** | **13.7** | **5.1** | **-3.3** | **-14.0** |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% | WB | | **-5.7** | **-5.1** | **2.0** | **11.7** | **16.7** | **22.8** | **25.2** | **23.4** | **18.7** | **11.3** | **1.4** | **-2.7** |
| (28) | | | MCDB | | **-5.0** | **-4.0** | **5.8** | **18.0** | **25.7** | **29.3** | **30.3** | **28.1** | **23.9** | **15.8** | **3.2** | **-2.0** |
| (29) | | 2% | WB | | **-9.1** | **-8.2** | **-0.4** | **8.3** | **14.6** | **20.5** | **23.0** | **22.0** | **17.2** | **8.8** | **-0.3** | **-7.1** |
| (30) | | | MCDB | | **-8.5** | **-6.9** | **2.8** | **14.4** | **22.5** | **26.7** | **28.9** | **26.5** | **21.4** | **11.9** | **0.9** | **-6.3** |
| (31) | | 5% | WB | | **-12.1** | **-10.3** | **-2.1** | **5.9** | **13.0** | **19.0** | **21.8** | **20.7** | **15.8** | **6.9** | **-1.7** | **-10.7** |
| (32) | | | MCDB | | **-11.5** | **-9.2** | **0.4** | **11.5** | **19.8** | **24.6** | **26.9** | **24.7** | **19.2** | **9.6** | **-0.4** | **-10.0** |
| (33) | | 10% | WB | | **-15.1** | **-12.2** | **-3.7** | **3.8** | **11.3** | **17.7** | **20.6** | **19.6** | **14.4** | **5.4** | **-3.3** | **-14.0** |
| (34) | | | MCDB | | **-14.6** | **-11.2** | **-1.6** | **8.4** | **17.3** | **23.0** | **25.3** | **23.2** | **17.5** | **8.1** | **-2.0** | **-13.4** |
| (35) | Mean Daily Temperature Range | | MDBR | | **11.7** | **13.6** | **13.1** | **10.5** | **11.8** | **11.9** | **10.4** | **9.7** | **10.9** | **9.3** | **9.5** | **10.6** |
| (36) | | 5% DB | MCDBR | | **8.7** | **12.9** | **14.3** | **15.4** | **17.6** | **15.5** | **14.0** | **13.0** | **14.0** | **12.8** | **8.9** | **8.4** |
| (37) | | | MCWBR | | **8.2** | **11.5** | **10.8** | **9.3** | **9.0** | **7.1** | **6.5** | **6.7** | **7.8** | **7.9** | **7.0** | **8.0** |
| (38) | | 5% WB | MCDBR | | **8.5** | **12.1** | **13.4** | **15.1** | **16.8** | **14.0** | **12.6** | **11.2** | **12.2** | **11.3** | **8.1** | **8.3** |
| (39) | | | MCWBR | | **8.1** | **10.9** | **10.2** | **9.4** | **9.2** | **7.1** | **6.6** | **6.3** | **7.1** | **7.4** | **6.5** | **7.9** |
| (40) | Clear-Sky Solar Irradiance | | taub | | **0.262** | **0.293** | **0.361** | **0.473** | **0.550** | **0.502** | **0.507** | **0.428** | **0.374** | **0.376** | **0.309** | **0.263** |
| (41) | | | taud | | **2.118** | **2.069** | **1.919** | **1.798** | **1.791** | **2.014** | **2.043** | **2.285** | **2.367** | **2.155** | **2.097** | **2.077** |
| (42) | | | Ebn at Noon | | **718** | **800** | **802** | **750** | **714** | **761** | **748** | **797** | **802** | **689** | **632** | **631** |
| (43) | | | Edh at Noon | | **72** | **103** | **149** | **192** | **204** | **165** | **158** | **117** | **96** | **96** | **74** | **63** |
| (44) | All-Sky Solar Radiation | | RadAvg | | **1.36** | **2.54** | **4.07** | **5.09** | **5.08** | **5.66** | **5.07** | **4.33** | **3.42** | **2.13** | **1.31** | **1.02** |
| (45) | | | RadStd | | **0.07** | **0.10** | **0.22** | **0.31** | **0.52** | **0.53** | **0.56** | **0.46** | **0.29** | **0.17** | **0.10** | **0.08** |

## Historical Trends

| | | DBAvg | Heating | | Cooling | | | Degree-Days | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 99% DB | 99% DP | 1% DB | 1% WB | 1% DP | HDD10.0 | HDD18.3 | CDD10.0 | CDD18.3 |
| | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) |
| (46) | Station Only | **N/A** | **N/A** | **-2.29** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** |
| (47) | Regional (8 neighbors) | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** |

Nomenclature: See separate page